Integration — Lesson 9 Test: Signed Area, Net vs Total

10 questions · show your work in the box · translate every negative answer into words (behind / lost / colder / less) · fractions welcome

Name: ______________________________ Date: ______________ Score: ______ / 10

1. A toy train runs backwards at a steady velocity of −4 meters per second for 3 seconds. Compute $\int_0^3 (-4)\,dt$ and say in words what it means. 1 pt

2. For $f(x) = -2x$ on $[0, 3]$: find the geometric SIZE of the triangle between the graph and the x-axis, then compute $\int_0^3 (-2x)\,dx$ with the FTC. Explain why the two numbers differ in sign. 1 pt

3. Evaluate $\int_0^2 (x - 5)\,dx$. Before computing, predict the SIGN of the answer from the graph and write your prediction down. 1 pt

4. Evaluate $\int_0^8 (x - 4)\, dx$ with the FTC, then explain the answer using two triangles. 1 pt

5. True or false, with a one-sentence explanation for each: (a) "A region's area can be negative." (b) "A definite integral can be negative." 1 pt

6. A cyclist's velocity is $v(t) = 6 - 3t$ miles per hour on $[0, 4]$. Find the NET displacement and the TOTAL distance ridden. (Hint: find where $v = 0$ first.) 1 pt

7. For $f(x) = x - 2$ on $[0, 5]$: find the crossing point, integrate the two pieces separately, then give the NET signed area and the TOTAL area of the regions. Fractions expected. 1 pt

8. A pool drains at 6 liters per minute for 4 minutes, then refills at 5 liters per minute for 2 minutes. (a) What is the net change in the pool's water? (b) How much water moved in total? 1 pt

9. Water flows in/out of a tank at the rate $r(t) = t^2 - 9$ liters per minute on $[0, 3]$. Evaluate $\int_0^3 (t^2 - 9)\,dt$ and interpret the answer — including why its sign was guaranteed on this interval. 1 pt

10. Find the number $b > 0$ that makes $\int_0^b (x - 3)\,dx = 0$, and explain in one sentence what is special about that moment. 1 pt